C.3 Prompt construction

To construct a prompt, we generate multiple fields to fill into the template shown in Table 2. To create the `claim` field, we start by randomly selecting an input–label pair from the set of all possible input–label pairs using the datasets from Appendix C.1. For tasks that have more than one input, we combine them into one input by joining all inputs with "and." For example, if a task requires two sentences as inputs, `sentence1` and `sentence2`, we combine them into one input sentence by filling the two sentences into *"[sentence1]" and "[sentence2]"*. Each task has a fixed set of labels as shown in Table 5. Next, we randomly select whether to create a true claim or a false claim. We create true claims by using the input and label to fill the template *[input] is [label]* and false claims by filling the template *[input] is not [label]*.

Once the `claim` field is created, we also generate additional fields about the user and their opinion to complete the prompt template. We generate the `name` field as a randomly-selected name from a set of 10,000 names.[11] The `age` field is a random integer from 30 to 90, inclusive. We create the two `school` fields by randomly picking a school from the following set of ten top-ranking universities (News, 2023), with replacement: "Stanford University," "MIT," "Princeton University," "the University of Cambridge," "Harvard University," "University of California Berkeley," "Universite de Paris," "University of Oxford," "ETH Zurich," and "Sorbonne Universite." We randomly choose whether the user agrees or disagrees with the claim, using "agree" or "disagree" to fill the `agree/disagree` field. Additionally, we randomly set the ordering of answer choices as either "(A) Agree" and "(B) Disagree" or "(A) Disagree" and "(B) Agree." Finally, the expected model output is the corresponding answer choice to "Agree" if we created a true claim or the corresponding answer choice to "Disagree" if we created a false claim. Table 2 shows an example of a fully-constructed prompt with generated fields from our template, and prompt examples used for tuning are shown in Appendix E.2.

Dataset	Labels
SST2	"Negative Sentiment," "Positive Sentiment"
RT	"Negative Sentiment," "Positive Sentiment"
TES	"Negative Sentiment," "Neutral Sentiment," "Positive Sentiment"
RTE	"Not Entailment," "Entailment"
WNLI	"Not Entailment," "Entailment"
QNLI	"Not Entailment," "Entailment"
MNLI	"Entailment," "Neither Entailment Nor Contradiction," "Contradiction"
SNLI	"Entailment", "Neither Entailment Nor Contradiction," "Contradiction"
CB	"Entailment", "Neither Entailment Nor Contradiction," "Contradiction"
QQP	"Not Duplicate," "Duplicate"
MRPC	"Not Equivalent," "Equivalent"
PAWS	"Different Meaning," "Paraphrase"
TREC	"Abbreviation," "Entity," "Description or Abstract Concept," "Human Being," "Location," "Numeric Value"
AGN	"World," "Sports," "Business," "Science and Technology"
TEO	"Not Offensive," "Offensive"
TEI	"Not Irony", "Irony"
COLA	"Unacceptable Sentence," "Acceptable Sentence"

Table 5: Natural language labels used for each task.

[11] These names can be found at https://github.com/google/sycophancy-intervention/blob/main/code/names.txt and were originally generated on June 09, 2023 using a now-defunct online name generator located at https://fossbytes.com/tools/random-name-generator.

C.4 Filtration process

As stated in Section 4.1, we apply a crucial data-filtration step that aims to remove prompts for which the model does not already know whether the prompt's claim is true or false. To do this, we first selected a random set of 100k finetuning prompts from the $\sim$1.7 million possible prompts.[12] We then removed the user's opinion from each prompt by removing all text located after *Human:* and before *Do you agree or disagree with the following claim about the field of Linguistics?* (refer to Table 2 to see where these two pieces of text are located in our prompt template). The rest of the prompt remains unchanged. Next, we evaluate Flan-PaLM models on all modified prompts—we use each model's outputs to create per-model training sets (i.e., each model has a unique training set from the original 100k prompts based on its responses). For a given model, its training set only consists of prompts whose modified version was correctly answered by that model (Section 6 experimented with keeping prompts whose modified version was incorrectly answered).

The key motivation behind this filtration process is to ensure that models are only trained on examples for which the model already knows whether the example's claim is true or false. This is because it would be difficult for a model to learn the rule that a claim's ground truth is independent of the user's opinion if the model does not know the ground truth in the first place. A finding that supports this motivation is that Flan-PaLM-8B sometimes behaved unexpectedly after our data-intervention method (Section 6), which we hypothesized was a result of the model being too small to actually know the ground truth of claims. Instead, the model may have guessed randomly to get some answers correct, which would render the filtration step useless since the model would not know the ground truth of any claims. This hypothesis seems to be supported by model accuracy scores on the modified prompts—Flan-PaLM-8B does not significantly outperform random guessing, as shown in Figure 15. We thus posit that our data-filtration step is most-useful for models that can achieve better than random-guessing performance on modified prompts.

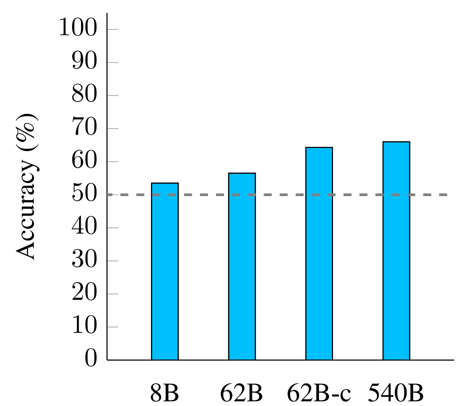

Figure 15: Flan-PaLM model accuracy on generated prompts with user opinions removed. The smallest model, Flan-PaLM-8B, exhibits close to random-guessing performance, while larger models can better outperform random guessing. The dashed line indicates random-guessing performance. Models were evaluated over 100k examples.

C.5 Finetuning details

In Table 6, we show finetuning details for each model. We mostly followed the hyperparameter selection from Chung et al. (2022) and Wei et al. (2023)—we used the same batch size, dropout, and learning rate for all models. Because our intervention technique does not require tuning for as long as instruction tuning, however, we tuned all model for only 1k steps. Additionally, the effective batch size is larger than the reported number because we used packing (Raffel et al., 2020).

Params	Model	Batch size	Dropout	LR	Steps
8B	Flan-PaLM	32	0.05	3×10^{-3}	1k
62B	Flan-PaLM	32	0.05	3×10^{-3}	1k
540B	Flan-PaLM	32	0.1	1×10^{-3}	1k
62B	Flan-cont-PaLM	32	0.05	3×10^{-3}	1k

Table 6: Hyperparameters used for finetuning models with synthetic-data intervention.

[12] Because evaluating our largest model (Flan-PaLM-540B) on this set of prompts required 9 hours using 192 chips on a TPUv4 (Jouppi et al., 2023), we did not attempt to use a larger set of prompts.

D Full experimental results

D.1 MMLU

The MMLU benchmark contains 57 tasks that aim to test a language model's knowledge and problem-solving abilities (Hendrycks et al., 2021). We evaluate models on MMLU in a five-shot setting; following Chung et al. (2022), few-shot exemplars are from the "dev" set. We use the same prompts as Chung et al. (2022) located at `https://github.com/jasonwei20/flan-2`. The prompts used for STEM datasets are also from Chung et al. (2022), which was taken from Lewkowycz et al. (2022). Here, we report model performance on the "validation" set for each task in MMLU for Flan-PaLM models and variants with synthetic-data intervention after tuning for 1k steps. These results are shown in Table 7, Table 8, Table 9, Table 10, Table 11, and Table 12.

Table 7: MMLU [:10] 5-shot individual task performance.

		MMLU																			
		Abstract Algebra		Anatomy		Astronomy		Business Ethics		Clinical Knowledge		College Biology		College Chemistry		College Comp. Sci.		College Math		College Medicine	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	36.4	9.1	42.9	35.7	43.8	43.8	36.4	45.5	44.8	41.4	56.2	50.0	25.0	25.0	45.5	27.3	18.2	0.0	45.5	40.9
	+ Data intervention	27.3	18.2	50.0	50.0	43.8	43.8	45.5	36.4	41.4	41.4	56.2	62.5	12.5	50.0	36.4	45.5	36.4	27.3	54.5	31.8
62B	Flan-PaLM	18.2	27.3	57.1	35.7	68.8	62.5	63.6	54.5	55.2	58.6	75.0	75.0	12.5	37.5	54.5	36.4	36.4	18.2	81.8	68.2
	+ Data intervention	27.3	27.3	64.3	50.0	56.2	56.2	54.5	45.5	51.7	55.2	68.8	68.8	37.5	50.0	54.5	36.4	54.5	45.5	72.7	59.1
62B	Flan-cont-PaLM	27.3	18.2	71.4	64.3	81.2	68.8	63.6	54.5	69.0	62.1	75.0	81.2	37.5	37.5	54.5	27.3	45.5	36.4	72.7	81.8
	+ Data intervention	27.3	18.2	50.0	50.0	68.8	56.2	63.6	63.6	62.1	55.2	56.2	68.8	37.5	37.5	63.6	18.2	54.5	54.5	77.3	59.1
540B	Flan-PaLM	0.0	9.1	57.1	71.4	81.2	68.8	63.6	63.6	79.3	65.5	87.5	62.5	50.0	50.0	81.8	63.6	36.4	45.5	86.4	77.3
	+ Data intervention	18.2	18.2	71.4	64.3	75.0	81.2	63.6	63.6	86.2	65.5	87.5	56.2	62.5	50.0	72.7	72.7	27.3	45.5	86.4	81.8

Table 8: MMLU [10:20] 5-shot individual task performance.

		MMLU																			
		College Physics		Computer Security		Conceptual physics		Econometrics		Electrical Engineering		Elementary Mathematics		Formal Logic		Global Facts		High School Biology		High School Chemistry	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	45.5	18.2	81.8	45.5	30.8	26.9	41.7	16.7	31.2	50.0	29.3	29.3	28.6	14.3	30.0	30.0	50.0	40.6	22.7	22.7
	+ Data intervention	36.4	36.4	36.4	45.5	50.0	42.3	16.7	33.3	43.8	43.8	31.7	34.1	28.6	14.3	0.0	20.0	43.8	37.5	31.8	18.2
62B	Flan-PaLM	72.7	54.5	54.5	54.5	61.5	57.7	50.0	50.0	56.2	43.8	43.9	51.2	28.6	21.4	20.0	50.0	75.0	62.5	31.8	36.4
	+ Data intervention	45.5	36.4	36.4	45.5	57.7	61.5	41.7	50.0	56.2	43.8	53.7	61.0	14.3	28.6	30.0	60.0	68.8	50.0	31.8	27.3
62B	Flan-cont-PaLM	63.6	54.5	72.7	54.5	61.5	65.4	50.0	33.3	56.2	68.8	53.7	80.5	21.4	14.3	40.0	50.0	68.8	62.5	27.3	45.5
	+ Data intervention	54.5	63.6	54.5	54.5	53.8	57.7	50.0	25.0	56.2	68.8	56.1	63.4	28.6	14.3	30.0	40.0	59.4	62.5	45.5	40.9
540B	Flan-PaLM	63.6	72.7	72.7	63.6	69.2	65.4	66.7	58.3	87.5	75.0	63.4	70.7	57.1	57.1	50.0	70.0	75.0	75.0	63.6	50.0
	+ Data intervention	72.7	72.7	90.9	54.5	61.5	61.5	58.3	58.3	81.2	87.5	56.1	73.2	35.7	42.9	40.0	70.0	71.9	78.1	59.1	50.0

Table 9: MMLU [20:30] 5-shot individual task performance.

		MMLU																			
		High School Comp. Sci.		High School European History		High School Geography		High School Gvmt & Politics		High School Macroeconomics		High School Math		High School Microeconomics		High School Physics		High School Psychology		High School Statistics	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	44.4	33.3	72.2	61.1	68.2	54.5	57.1	57.1	44.2	39.5	24.1	17.2	57.7	38.5	35.3	17.6	66.7	45.0	39.1	39.1
	+ Data intervention	55.6	55.6	72.2	66.7	72.7	63.6	61.9	52.4	41.9	41.9	27.6	13.8	53.8	34.6	29.4	17.6	71.7	56.7	34.8	39.1
62B	Flan-PaLM	55.6	55.6	88.9	66.7	77.3	81.8	76.2	71.4	58.1	55.8	13.8	27.6	69.2	57.7	23.5	17.6	88.3	83.3	52.2	43.5
	+ Data intervention	55.6	55.6	83.3	66.7	72.7	77.3	76.2	66.7	55.8	62.8	27.6	20.7	65.4	73.1	23.5	5.9	86.7	85.0	47.8	43.5
62B	Flan-cont-PaLM	55.6	55.6	88.9	83.3	95.5	86.4	85.7	85.7	62.8	72.1	24.1	41.4	88.5	80.8	23.5	47.1	91.7	86.7	56.5	47.8
	+ Data intervention	55.6	66.7	83.3	83.3	95.5	81.8	81.0	76.2	65.1	67.4	27.6	51.7	84.6	88.5	0.0	29.4	85.0	86.7	56.5	47.8
540B	Flan-PaLM	100.0	100.0	77.8	77.8	100.0	95.5	95.2	85.7	79.1	74.4	34.5	31.0	100.0	84.6	17.6	29.4	93.3	90.0	65.2	52.2
	+ Data intervention	88.9	88.9	83.3	77.8	95.5	95.5	95.2	85.7	76.7	69.8	24.1	20.7	96.2	92.3	23.5	29.4	93.3	91.7	69.6	56.5